



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch : III Semester B.E CSE
Course Code & Title : CS3352 & Foundations of Data Science
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit II/ Describing Data/Describing Data with Averages
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TL04
- **Activity Chosen:** Problem Based Learning
- **Justification**
 - ❖ Problem-based learning is an educational approach in which the students learn by solving some problems and understand the core concepts clearly.
 - ❖ Describing data with averages is an evaluation in the finding the mean, median and frequency distribution. To better understand the error calculation process, the students are made to solve a problem.
 - ❖ The prime objective of problem based learning is to enhance the problem solving and creative thinking of the students.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
 - ❖ Faculty explained the problem and formulas of Relative Frequency Distribution, Cumulative Frequency Distribution and Describing Data prior to the activity.
 - ❖ Faculty divided students into teams. Total number of team is 17. Number of members in each team is 3.
 - ❖ Then faculty gave problem on describing data with averages.
 - ❖ Student teams solved the problem with calculation steps with their partners.
 - ❖ Each member of Ms. R. Sathya Pushpa came in front to explain the concepts clearly.
 - ❖ Ms. A. Annalakshmi of this class solved the relative frequency Distribution and explained the step by step solution to entire class.
 - ❖ Ms.B.Sivasundari solved the Cumulative Frequency Distribution and explained the step by step solution to entire class.
 - ❖ Ms. R. Sathya Pushpa also solved the Describing Data and explained the step by step solution to entire class.
 - ❖ Finally, the instructor consolidated the information that was discussed in this activity and related problems explained on the board.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO1
CO1	3	3	3	1	1	1	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovation Practice	PO1	PO2	PO3	PO10	P012	PSO1
	2	2	3	2	2	3
Justification for Correlation	Apply basic knowledge and fundamentals of statistical techniques and describe the data.	Identify the need for statistical techniques and can describe the data for any applications.	Able to design solutions for any applications and perform statistical descriptions of data	Ability to recognize the need for statistical techniques for describing data.	Interpret statistical descriptions of data and prepare as a report or presentation	Exhibit knowledge in statistical descriptions of data for application development.

- **Images / Screenshot of the practice:**



Figure: Students actively solving problems on the board

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***

Students enjoyed in solving the problem as a team and effectively competed with other student members.

❖ ***Feedback of practice from students and other stakeholders:***

- The students felt that the problem based learning on subject related problems are interesting than usual theory classes.
- This activity encouraged the students to share their knowledge with others.

❖ ***Benefit of the practice:***

- Students learned a few things individually that leads to self-learning. They also actively involved in this session rather than it being a one-way communication.
- Students were able to connect ideas through discussion.

❖ ***Challenges faced in implementation***

- Students with a slower learning pace initially faced challenges in solving problems independently due to their individual learning backgrounds. Nevertheless, they grasped the concepts effectively through active participation and interaction within the team.

References:

- ❖ <http://www.et.iitb.ac.in/TeachingResources.html>
- ❖ <https://www.hunschool.org/resources/problem-based-learning>
- ❖ [https://teaching.cornell.edu/teaching-resources/engaging-students/problem-basedlearning 2023/2.KVS_TPS_FDS.pdf](https://teaching.cornell.edu/teaching-resources/engaging-students/problem-basedlearning%2023/2.KVS_TPS_FDS.pdf)
- ❖ [https://citl.illinois.edu/citl-101/teaching-learning/resources/teachingstrategies/problem-based-learning-\(pbl\)](https://citl.illinois.edu/citl-101/teaching-learning/resources/teachingstrategies/problem-based-learning-(pbl))

Signature of Faculty Member

HOD